

Ex. No. : 02

Date:

Register No.: 231701014

Name: N GOKUL KRISHNA

A PYTHON PROGRAM TO IMPLEMENT SIMPLE LINEAR REGRESSION USING LEAST SQUARE METHOD

Aim:

To implement **Simple Linear Regression** using the **Least Square Method** in Python and analyze the relationship between one independent variable and one dependent variable from the Diabetes dataset.

Requirements:

Jupyter Notebook with Python, Diabetes.csv dataset, and required Python libraries.

PROGRAM:

```
import pandas as pd

# Load dataset from your folder
from sklearn.datasets import
load_diabetes

# View first 5 rows
print(df.head())

print(df.info())

print(df.describe())

X = df[['bmi', 'bp']]

y = df['target']
```

```
from sklearn.model_selection import train_test_split
```

```
X_train, X_test, y_train, y_test = train_test_split(  
    X, y, test_size=0.2, random_state=42)
```

```
from sklearn.linear_model import LinearRegression
```

```
model = LinearRegression()
```

```
model.fit(X_train, y_train)
```

```
print("Intercept:", model.intercept_)
```

```
print("Coefficients:", model.coef_)
```

#Output:

```
Intercept: 151.8153925749251
```

```
Coefficients: [839.51985334 393.45120312]
```

```
y_pred = model.predict(X_test)
```

```
from sklearn.metrics import mean_squared_error, r2_score
```

```
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
```

```
print("R2 Score:", r2_score(y_test, y_pred))
```

#Output:

```
Mean Squared Error: 3740.5070199204715
```

```
R2 Score: 0.2939977566766082
```

```
import matplotlib.pyplot as plt

plt.scatter(X_test['bmi'], y_test)

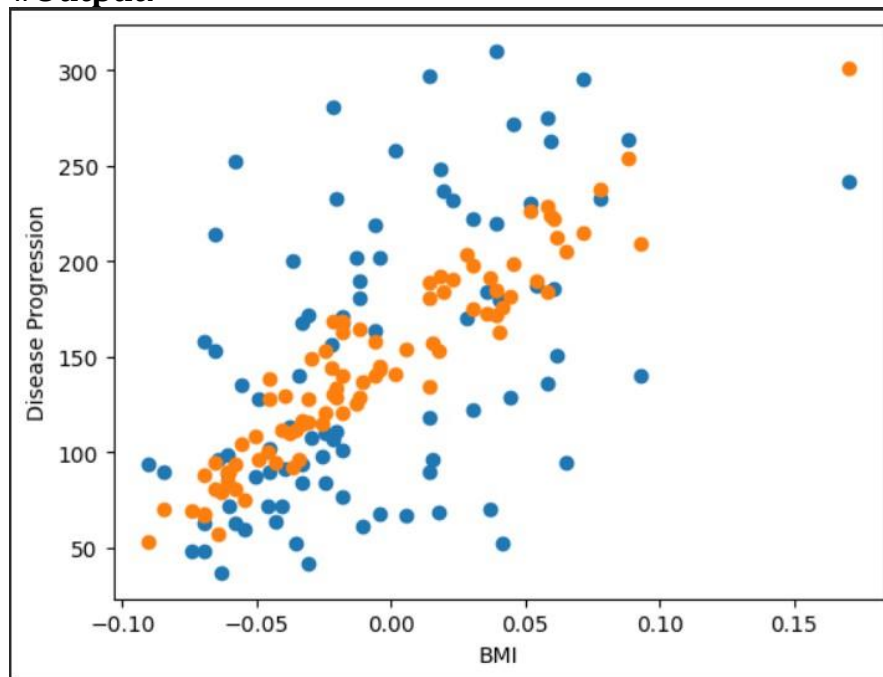
plt.scatter(X_test['bmi'], y_pred)

plt.xlabel("BMI")

plt.ylabel("Disease Progression")

plt.show()
```

#Output:



#Result:

The Simple Linear Regression model was successfully implemented on the Iris dataset. The Least Square Method was used to calculate the best-fit regression line. The model was able to predict the dependent variable with reasonable accuracy, showing a linear relationship between the selected features. The performance was evaluated using metrics such as Mean Squared Error (MSE) and R^2 score.